A low degree of heterogeneity in vehicle speeds can lead to large-scale change of mobility for urban traffic dynamics

N. ABDUL MAJITH[1] and Sitabhra SINHA[1,2]

[1]The Institute of Mathematical Sciences, CIT Campus, Taramani, Chennai 600113, India
[2]National Institute of Advanced Studies, IISc Campus, Bangalore 560012, India.

E-mail: [1]majithabdul@imsc.res.in, [2]sitabhra@imsc.res.in

Abstract

Vehicular traffic provides a fascinating perspective into large-scale social dynamics where the micro-dynamics of interactions between self-propelled agents results in macroscopic changes of phase, e.g., from free flow to jammed states. Here we use cellular-automata based models of traffic dynamics to study the role of vehicle heterogeneity (i.e., the occurrence of slow and fast moving vehicles in the same road) on the transition in mobility as the rate at which vehicles enter the road is varied. We focus on the specific case of flow across a traffic intersection involving two perpendicular thoroughfares. A signal having a duty cycle with a given period is located at the intersection which periodically allows traffic to flow along one road while that along the perpendicular road is stopped. In this characteristic urban situation, we find that a very small proportion of slow moving vehicles disproportionately affect the overall mobility by reducing it markedly. We also observe that increasing the number of lanes while keeping the overall vehicle density constant, results in small changes in the critical value of entry flow volume per lane at which transition from free flow to jamming occurs; however, the mobility in free flow region is seen to increase with the number of lanes. We also observe that lane indiscipline tends to decrease mobility in the free flow region but does not appear to affect the transition to jamming.

Keyword: Social simulations, Urban traffic dynamics, Cellular automata, Vehicle heterogeneity

The behavior of vehicular traffic provides one of the more amenable arenas for studying social dynamics where large number of agents interacting with each other produces unexpected collective phenomena [1]. In particular, urban traffic characterized by high densities, variable flow patterns and signals coordinating the movement of vehicles converging from several directions, is a challenging area for theoretical and simulation studies. Previous work on highway traffic and to a lesser extent, urban traffic, has revealed several robust features, such as the transition from free flow to jammed state at a critical density (see e.g., Ref. [2]). In this work, we investigate the role of heterogeneity in the average speed at which vehicles move on the overall mobility of traffic. We use a cellular automata modeling framework, with a thoroughfare being represented as a lattice of discrete sites, vehicles moving at a fixed speed from one site to another - as long as the site they wish to occupy is empty. If another vehicle occupies the site, the vehicle cannot complete the move. For simplicity, we consider two classes of vehicle, one capable of moving at twice the speed of the other (whose speed is taken to be unity, by suitable choice of units). In particular, we consider the situation around a traffic intersection comprising two perpendicular roads with vehicles moving along each in a particular direction (Fig. 1).

A traffic signal placed at the intersection coordinates the flow of vehicles moving in the perpendicular roads. The signal allows vehicles in a particular road to move (causing vehicles in the other direction to stop when they arrive at the intersection) for one half of a duty cycle. Vehicles enter each road from outside with a rate np_{entry}, where n is the number of lanes and p_{entry} characterizes the entry flow volume per lane. Another important parameter is the fraction of vehicles having slower speed f_{slow} ($0 \leq f_{slow} \leq 1$). For each set of parameter values, we measure the overall mobility of the system (i.e., weighted average of the move made by each vehicle). We notice that at high values of p_{entry}, as f_{slow} increases from 0 to 0.1 there is a drastic drop in mobility from 1 to a low value. With further increase in f_{slow} the mobility starts increasing again. This non-monotonic behavior of mobility is not observed for lower values of p_{entry}, pointing to an intriguing collective phenomenon of heterogeneous traffic dynamics. We explain how

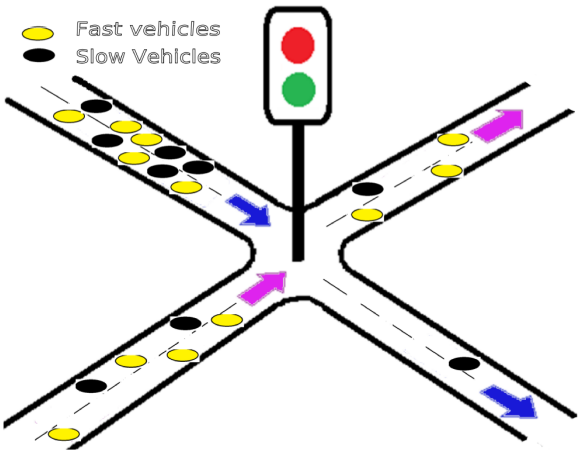


Figure1: Schematic representation of traffic dynamics around an intersection with a signal. The colored circles on the roads represent vehicles moving in the direction indicated by the arrows, with colors corresponding to the speed at which they can move. The traffic signal (indicated by a post at the intersection) coordinates the traffic so that vehicles along one direction is stopped for one half of a duty cycle while those in the perpendicular direction move.

---

this non-intuitive situation comes about and show the effect of increasing the number of lanes on this behavior. We also show that lane indiscipline while decreasing mobility in the free flow region does not appear to affect the transition to jamming. Our results provide new insights into how the coexistence of vehicles with widely differing speeds (as is commonly seen in the cities of developing countries) can affect traffic flow patterns.

References

[1] P. Ball, "The physical modelling of human social systems", Complexus vol. 1, 190 (2003).

[2] O. Biham, A. Middleton and D. Levine, "Self-organization and a dynamical transition in traffic-flow models", Physical Review A, vol. 46, 6124 (1992).